

CANDIDATE NAME: _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

H2 BIOLOGY

Paper 3 Long Structured and Free-response Questions

9744/03

Candidates answer on the Question Paper.
No Additional Materials are required.

Tuesday
18 September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CG in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
1	/27
2	/23
Section B	/25
Total	/75

Section A

Answer **all** the questions in this section.

Question 1

(a) Outline the features found within a typical bacterium cell that distinguishes it from a typical plant cell. [3]

1. Ref to **lack of membrane-bound organelles** in bacteria vs. **presence** of such organelles in plant cells
2. Ref to bacterial cells containing **circular DNA** vs. plant cells containing **linear DNA**
3. Ref to DNA of bacterial cells found in **nucleoid** region vs. DNA in plant cells found in **nucleus**
4. Ref to bacterial cells containing **70S ribosomes** vs. plant cells containing **80S ribosomes** in the cytoplasm
5. Ref to bacterial cells containing **plasmids/ extra-chromosomal DNA**, which are **absent** in plant cells
6. Accept AVP

Max 3

Reject: comparisons involving cell wall and flagella.

Ignore: comparisons involving chloroplasts, vacuoles, mitochondria etc.

(b) One other way that bacteria differ from plants is in terms of how their genome is organised.

In bacteria, genes are generally organised into operons. Explain the advantage of this arrangement. [2]

1. The clustering of **several genes** that are **functionally related** (*or give idea of*) under a **single promoter**
2. Allows for **co-ordinated/ common regulation** or ref. to idea of **efficient gene expression**
3. Ref. to operons contributing to **compact chromosome size** in bacteria due to reduced need for multiple promoters etc.
4. Accept AVP

Max 2

(c) Like plant cells, bacterial cells are also surrounded by a rigid cell wall. However, while cellulose is a major component in plant cell walls, the significant constituent of the bacterial cell wall is peptidoglycan.

The peptidoglycan in some bacterial cell walls has the following features:

- It consists of linear chains of two alternating amino-sugars, N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM).
- In both NAG and NAM, the sugar component is covalently attached to a short oligopeptide sequence containing 4 to 5 amino acid residues.
- D-Alanine (D-Ala) and D-glutamine (D-Glu) are examples of amino acids that are commonly found in these sequences.

The structure of peptidoglycan in such bacterial cell walls is shown in Fig. 1.1 below.

A close-up of Region X is shown in Fig. 1.2 on page 4.

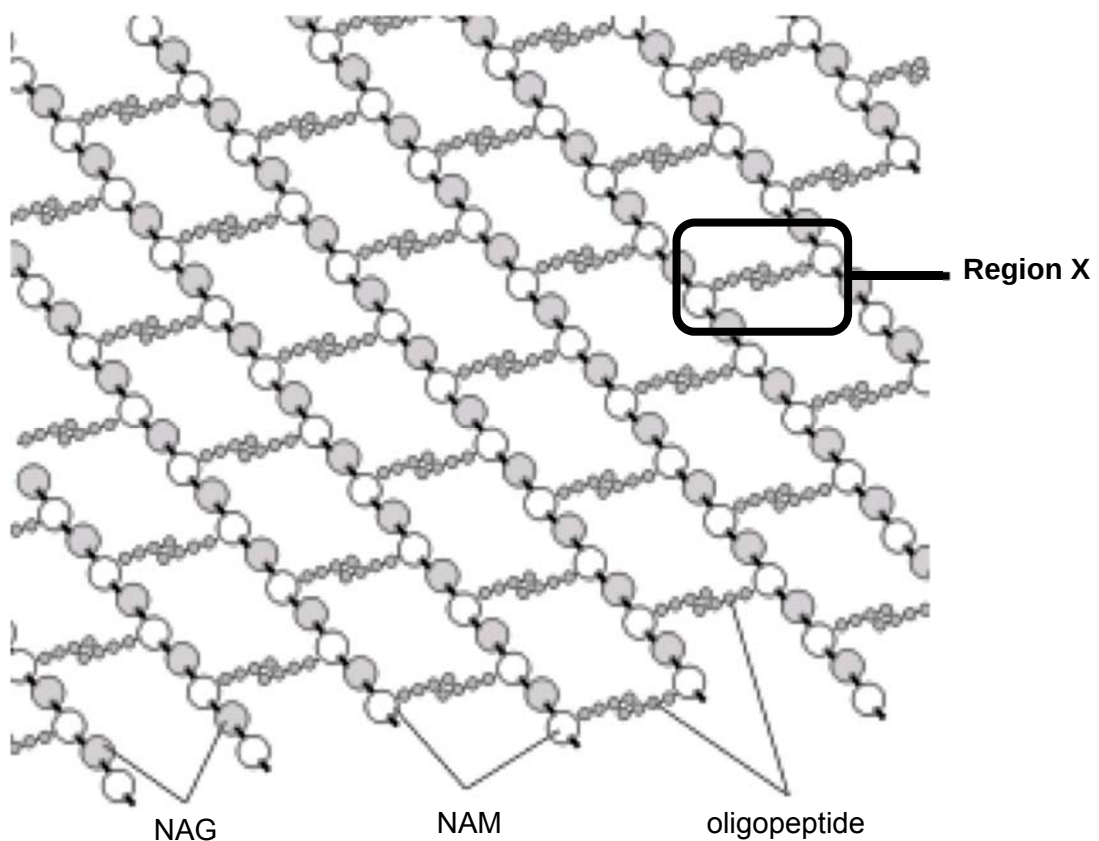
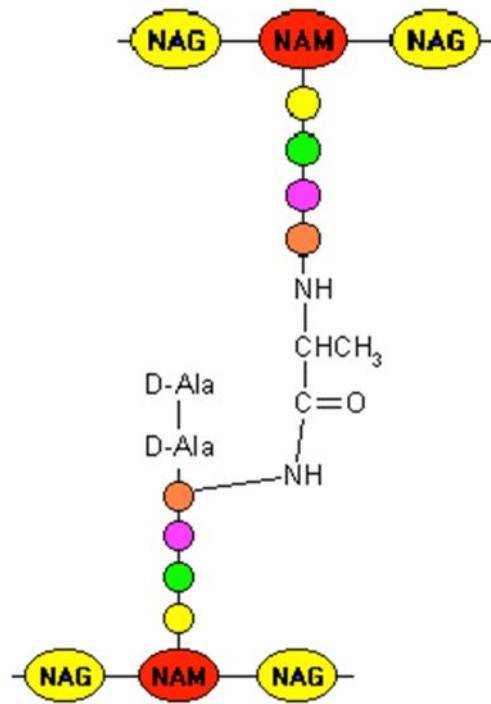


Fig. 1.1



Close-up of Region X

Fig. 1.2

(i) Describe how specific chemical groups of amino acids such as D-Ala and D-Glu allow them to be incorporated as part of an oligopeptide. [3]

1. Ref. to the presence of an **amino group** and a **carboxyl group** in these amino acids
2. Which form **peptide bonds** with other amino acids
3. In **condensation reactions** with a **water molecule lost** with each peptide bond formed

(ii) Like cellulose, extensive cross-links are present in peptidoglycan.

With reference to the given information including Figs. 1.1 and 1.2, distinguish between the cross-links in cellulose and peptidoglycan. [2]

1. In peptidoglycan, cross-links are formed **between oligopeptide chains** of NAM residues, while cross-links are formed **between –OH groups** of β -glucose residues in cellulose.
2. In peptidoglycan, cross-links are in the form of **peptide bonds**, while cross-links are in the form of **hydrogen bonds** in cellulose.

(iii) Suggest how the plant cell wall and bacterial cell wall are similar in function. [1]

1. **Maintains the shape** of the cell/ gives cell its **regular shape**/ provide **structural support**
2. Ref. to **protective** function
3. Ref. to preventing cells from bursting/ lysing due to influx of water

Max 1

- (d) The formation of cross-links in peptidoglycan is catalysed by penicillin binding protein (PBP). A common antibiotic which targets this enzyme is penicillin. By inhibiting PBP, penicillin thus prevents the synthesis of the cell wall.

Fig. 1.3 below is a simplified sequence of events showing the action of PBP (1 – 3), and the action of penicillin on PBP (4 – 5).

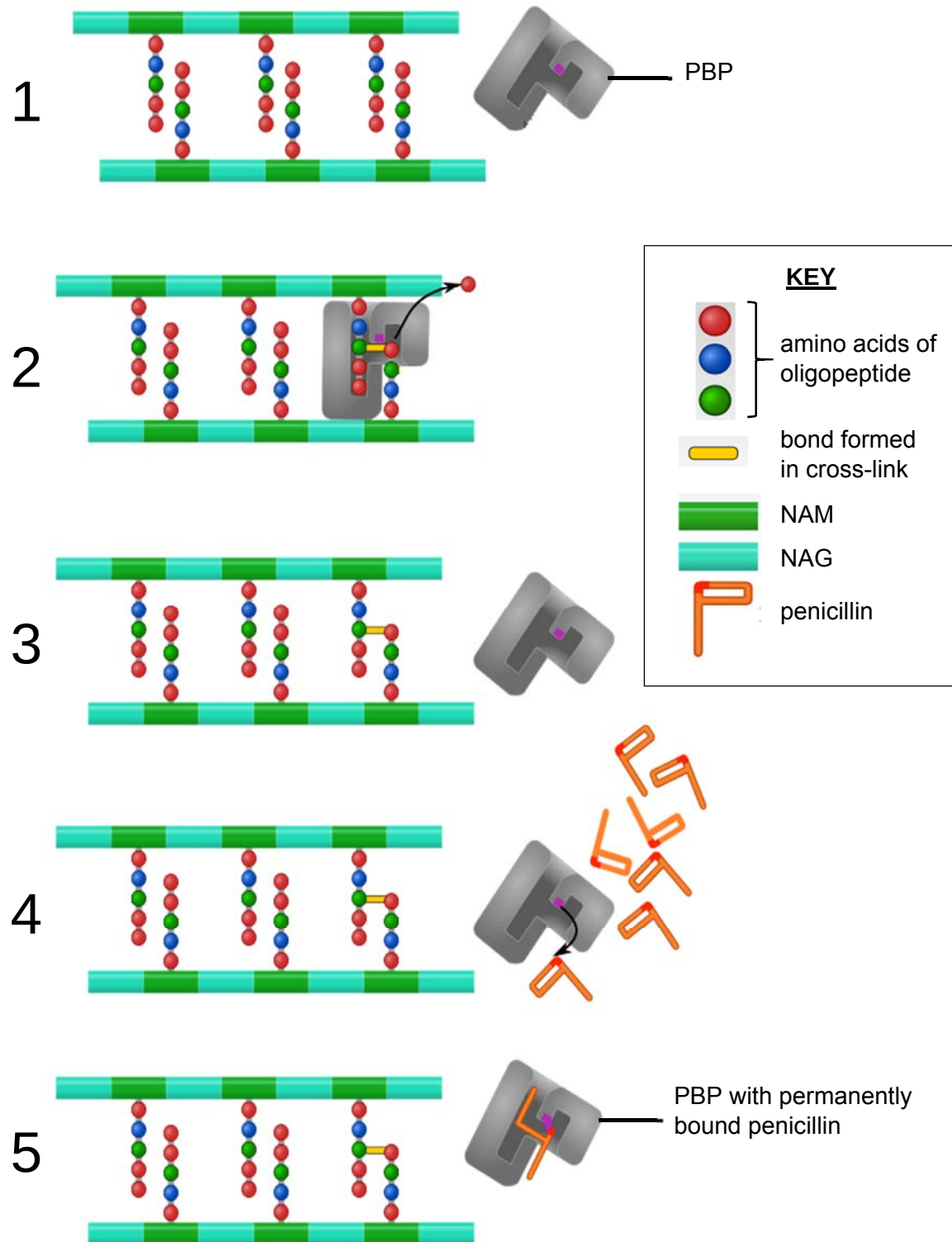


Fig. 1.3

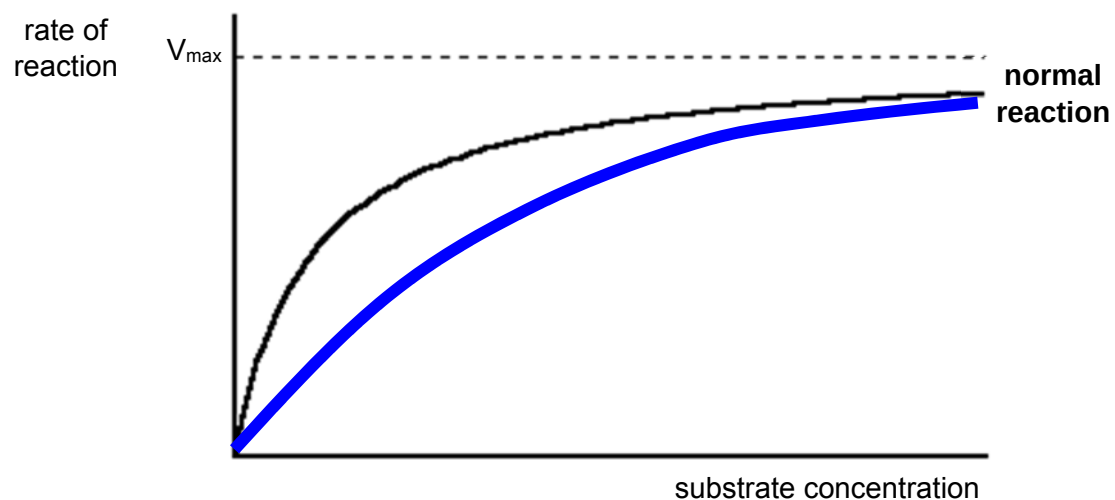
- (i) Although penicillin behaves like a typical competitive inhibitor, it is **not** a competitive inhibitor.

Compare the mode of action of penicillin with that of a typical competitive inhibitor. [2]

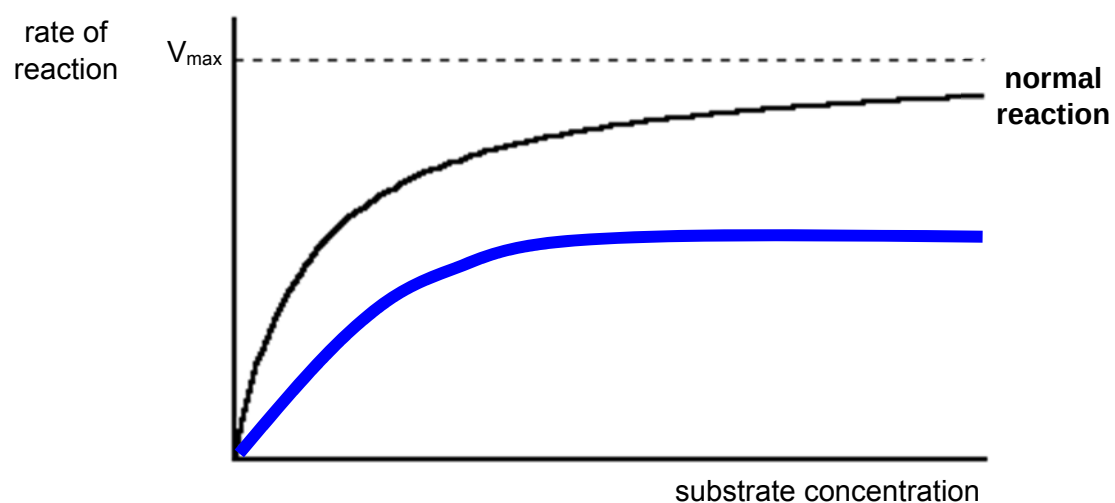
1. Similarity: Both penicillin and a typical competitive inhibitor bind to the active site of the enzymes that they inhibit. (*Reject: similarity in shape to substrate*)
2. Difference: Penicillin binds **irreversibly/ permanently** to penicillin-binding protein, while a typical competitive inhibitor **always binds reversibly** to its enzyme.

- (ii) By drawing an appropriately labelled curve each, complete the graphs below to show the effect of

- a typical competitive inhibitor on its enzyme. [1]



- penicillin on PBP. [1]



- (e) Another recently discovered antibiotic, SRJC30, inhibits bacterial growth by inhibiting protein synthesis.

SRJC30 functions by binding to the bacterial ribosome as shown in Fig. 1.4 below.

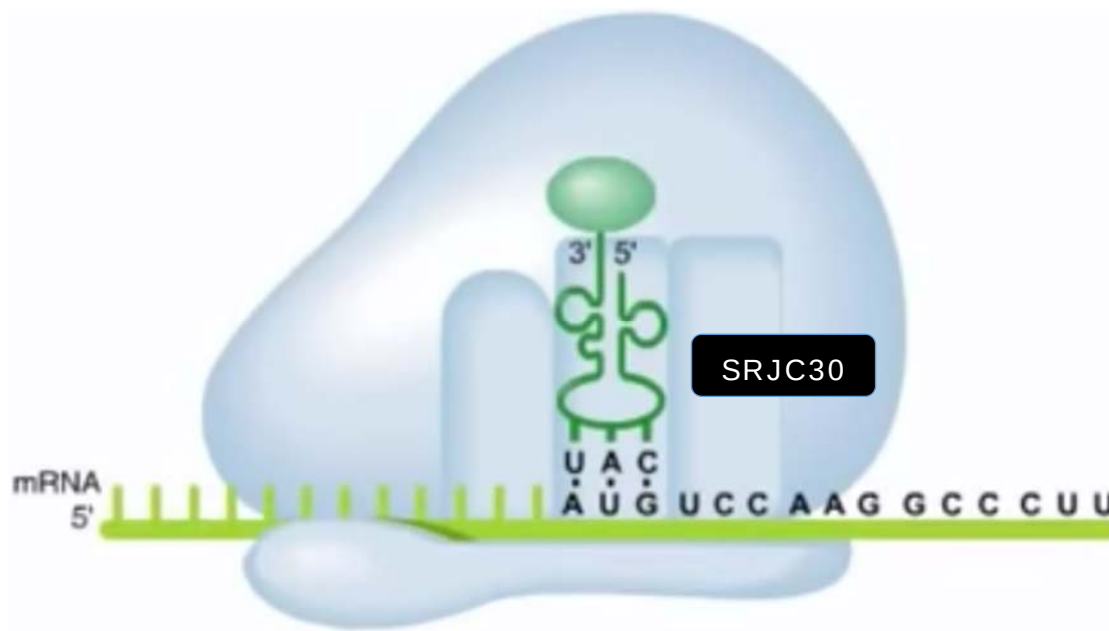


Fig. 1.4

Describe fully how SRJC30 inhibits protein synthesis.

In your answer, highlight **based on the context in Fig. 1.4**,

- the relevant specific locations on the ribosome and
- the specific molecules whose roles in protein synthesis have been prevented by SRJC30. [4]

1. SRJC30 binds to the **A site** on the **large/ 50S ribosomal subunit**
2. Thus **preventing** the **tRNA with the anti-codon AGG** and with the next amino acid from entering the A site
3. and undergoing **complementary base pairing** with the **UCC codon** on the mRNA at the A site
4. **Peptidyl transferase** on the large ribosomal subunit is thus **unable to catalyse the formation of a peptide bond**
5. Between **formyl-methionine** held by the initiator **tRNA at the P site** with this incoming amino acid.

Max 4

- (f) In a bacterial infection, the invading pathogen is recognised by the host's immune system as non-self. A co-ordinated and specific immune response is then mounted against it.

However, in treatment, the administered antibiotic is not recognised by the host's immune system, even when it is present in the bloodstream.

In modern medicine, antibiotics thus complement the host's immune system in fighting off a bacterial infection.

- (i) Outline how a co-ordinated and specific immune response against a bacterial pathogen is possible.

In your answer, highlight the role of the specific immune cell that is directly involved in co-ordinating this response. [4]

1. Activated **CD4 T helper cells** are the cells that co-ordinate a specific adaptive immune response to bacterial pathogen

Max 3 from any of the following:

2. Naïve CD4 T helper cells, after binding via their **T-cell receptors** to **peptide:class II MHC complexes** of **antigen presenting cells** are **activated by the cytokines** (*credit once only*) released
3. Before binding to **mature naïve B cells** and **releasing cytokines**, which causes these B cells to differentiate and proliferate into **plasma B cells** and **memory B cells** (humoral response).
4. plasma cells secrete **antibodies** that can cause **neutralisation/ opsonisation/ complement activation** (*any one*) against bacteria (*ignore: antibody dependent cell-mediated cytotoxicity*)
5. **Cytokine release** (*credit once only*) by these activated T helper cells also causes **naïve CD8 T cells** to proliferate and differentiate into **cytotoxic T cells** (cell-mediated response).
6. T cells secrete **cytotoxic proteins** (or give **named example** e.g. perforin, granulysin, granzymes) that **kill infected cells**.

- (ii) Suggest why the host's immune system does not recognise an administered antibiotic. [1]

1. Unlike antigens, antibiotics are **not protein** (or polysaccharide) in nature
2. Ref. to **small size** of antibiotic molecules

Max 1

- (g) Antibiotic resistance among bacteria is a growing global problem. Recent evidence indicates a relationship between antibiotic resistance and climate change.

Fig. 1.5 presents a finding from a study conducted in the United States, and shows a scatter plot of resistance among *E.coli* against the antibiotic amoxicillin, against minimum temperature.

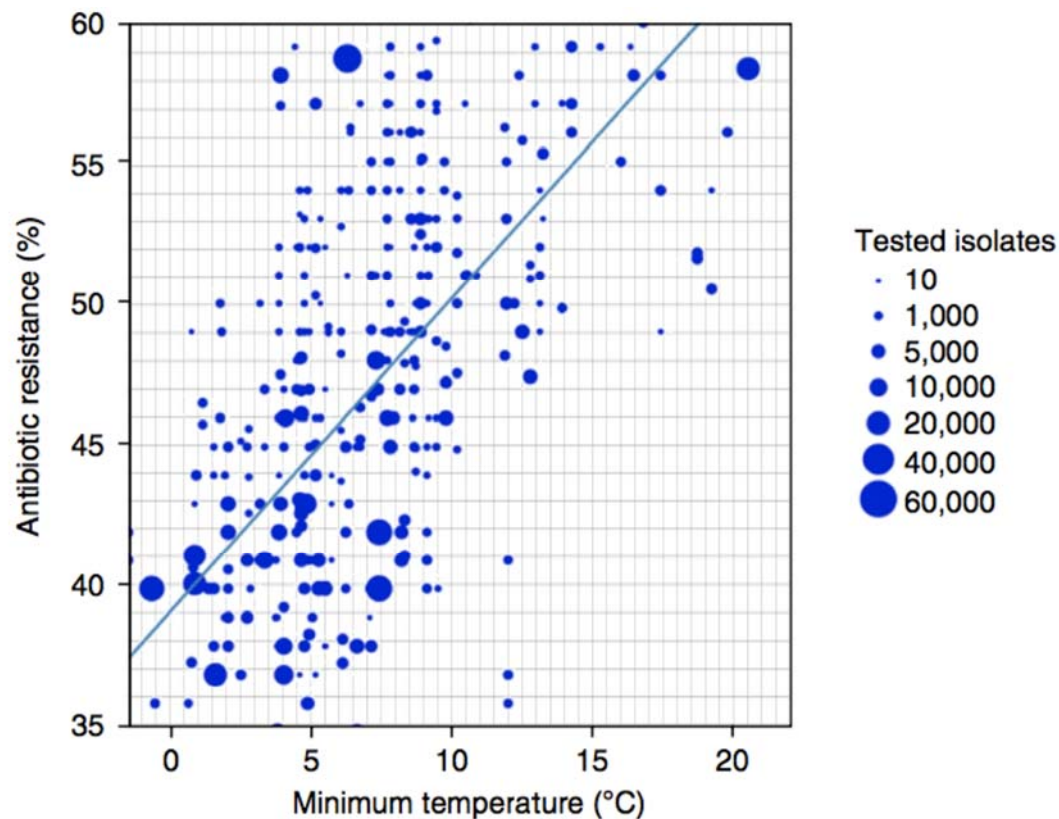


Fig. 1.5

Comment on how amoxicillin resistance among *E. coli* is affected with every 10 °C rise in minimum temperature, and suggest **two** reasons for this relationship. [3]

1. Amoxicillin resistance **increases** by **12%** (A: 11 or 13%) with every 10 °C rise in minimum temperature
2. Ref. to increasing temperature **facilitating horizontal gene transfer/ conjugation/ transduction/ transformation/ uptake of free DNA**
3. Ref. to increasing temperature **increasing the rate of reproduction via binary fission**

[Total: 27]

Question 2

Barley yellow dwarf virus (BYDV) is a positive sense single-stranded RNA virus; the virion is not enveloped in a lipid coating. The virus is transmitted by aphids, and the taxonomy of the virus is based on genome organisation, serotype differences and on the primary aphid vector of each isolate.

- (a) Explain why viruses may be considered both living and non-living. [2]
1. They have genome/ genetic material/ nucleic acid (Reject RNA/DNA)
 2. But no cytoplasm and organelles/ require host cell metabolic machinery for replication/ mention one or more enzymes for specific function
- (b) Using the information above and your knowledge, list two main classes of biomolecules present in a BYDV virion. [1]
1. Nucleic acid (Reject DNA/ RNA/ nucleotides)
 2. Proteins
Reject carbohydrates
- (c) (i) Explain briefly how the virus is able to produce a complete new virion using the starting material of a positive sense single stranded RNA. [3]
1. Translation of positive sense single stranded RNA into viral proteins
 2. Assembly of proteins into a viral capsid
 3. Positive RNA used as template to make complementary negative RNA, which is in turn used as a template to make positive RNA genome / Positive RNA used as template to make complementary DNA via reverse transcription, which is in turn used to synthesize complementary strand to form double-stranded DNA that is transcribed to form + RNA genome
 4. Packaged into viral capsid (Reject: Nucleocapsid)

Max 3

- (c) (ii) Suggest one enzyme that is present in its host that the virus would need for the process explained in (i), and one other enzyme that the host is unable to provide. [2]
1. Peptidyl transferase
 2. RNA dependent RNA polymerase
- Mark for ECF from (c)(i)

The symptoms of a BYDV infection vary with the age of the plant at the time of infection, the strain of the virus and the environmental conditions. Symptoms appear approximately 14 days after infection. Affected plants may show a yellowing or reddening of leaves, stunting, an upright posture of thickened stiff leaves, reduced root growth, delayed (or no) heading, and a reduction in yield. Young plants are the most susceptible, and infected wheat leaves have a reduced ability to photosynthesise.

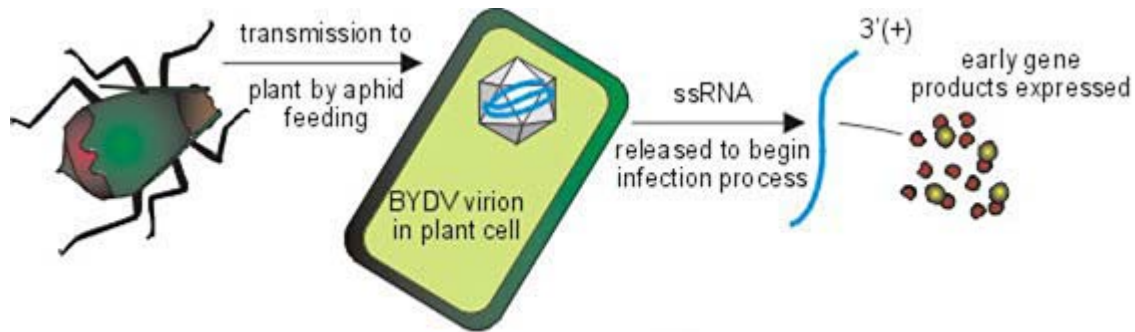
- (d) Explain briefly how disrupting photosynthesis may lead to stunted growth. [3]
1. No ATP/carbohydrates/starch/glucose made (effect of disrupting photosynthesis)
Reject food/ chemical energy/ nutrients
 2. Unable to grow (link between effect and symptom)

The host range of BYDVs consists of more than 150 grass species in the family *Poaceae*. A large number of grasses both annual and perennial are alternate hosts to the BYDV and can serve as reservoirs of the virus.

There are two main sources by which a cereal crop might be infected:

1. By non-migrant wingless aphids already present in the field and which colonise newly-emerging crops. This is known as "green-bridge transfer".
2. By winged aphids migrating into crops from elsewhere. These then reproduce and the offspring spread to neighbouring plants.

Transmission from an aphid is demonstrated in Fig. 2.1 below.



- (e) With reference to the grasses that serve as alternate hosts for the virus, explain what you understand about the concept of a species. [2]
1. Ecological species concept where different species (of aphids) occupy different niche
 2. The different grasses help to define the different niches of the different (aphid) species.
- OR
1. Biological species concept where the grasses are classified at 150 different species
 2. Unable to produce fertile viable offspring
- (f) Climate change has affected the life cycle and distribution of animals and plants in the world. Discuss how climate change may impact the rates of BYDV infection in a **localised** crop population. [5]
1. Increase in global temperature leads to increase in reproduction rate/AVP of aphid vectors.
 2. Due to increased rate of enzyme reactions involved in metabolic processes.
 3. Therefore, increase in aphid population leading to increased infection rates
 4. Climate change may cause migration of aphid vectors northwards to cooler climates.
 5. Leads to a drop in aphid populations at the location of the crop leading to decreased infection rates.
 6. Migration of grasses that serve as alternate aphid hosts northwards to cooler climates
 7. Leads to a drop in aphid populations at the location of the crop leading to decreased infection rates.
 8. AVP (Links climate change to aphid/ viral infection rates and is well explained/ substantial logical explanation of plant life cycle being affected by climate change and its relation to vector or viral population)

Max 5

- (g) Many farmers in rural farms in third world countries are struggling to cope with the impact of pests and pathogens such as the BYDV. Suggest why many of these farmers are unable to overcome these issues. [1]
1. Lack of finance to buy new types of seeds (e.g. pest-resistant seeds) **Reject: afford solutions/ lack resources/ lack money without clear explanations**

2. Lack of skills to branch into alternative farming techniques (e.g. start growing new crops not affected by the pests)
3. AVP

The following is an excerpt from a research paper on BYDV infections of two types of grass.

The wheat–Thinopyrum intermedium translocation line YW642 carries BYDV resistance gene BVDV-CP. To explore resistant wheat resistance in response to BYDV infection, we used GeneChip® Wheat Genome Arrays to analyze transcriptomes of YW642 and its susceptible parent Zhong8601 at 12 and 72 h postinoculation with BYDV.

Fig. 2.2 below shows the result of this investigation.

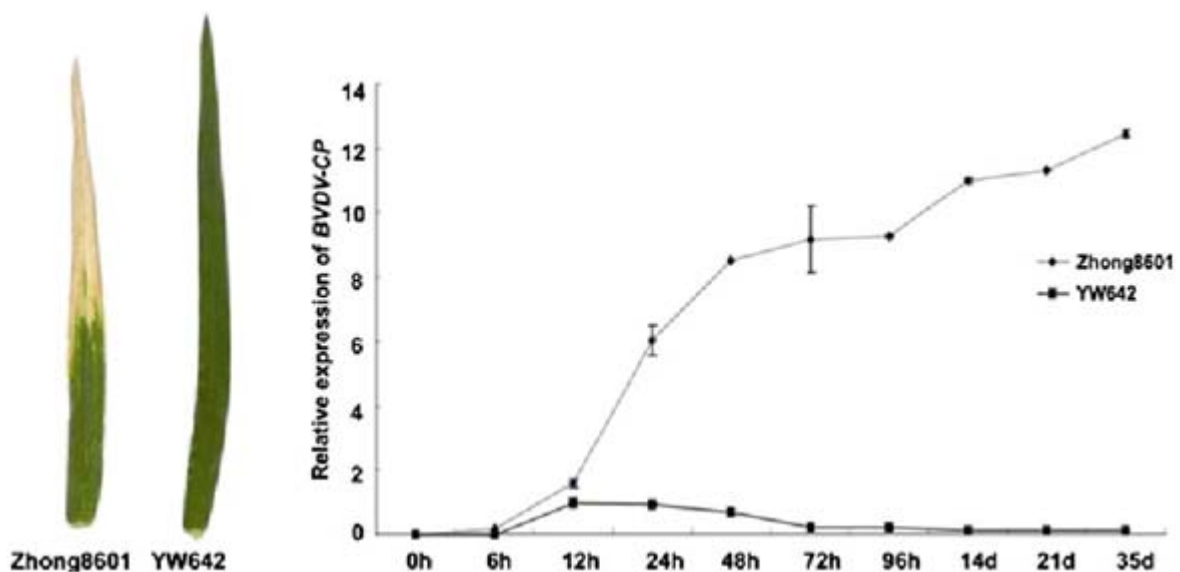


Fig. 2.2

- (h) Using the information and data provided, describe the effects of inoculating the two grass strains with BYDV on the expression of the resistance gene *BVDV-CP*. [2]
1. In YW642, relative expression of *BVDV-CP* remains 0 from 0 to 6h before increasing to 1 at 12h and decreasing to 0 at 14days
 2. In Zhong8601, relative expression of *BVDV-CP* remains 0 from 0 to 6h before increasing to 12.5 (reject 14) at 35 days
- (i) Suggest why the relative expression of the resistance gene *BVDV-CP* was low for both strains at 6 hours. [1]
1. Time needed for gene to be switched and the gene products expressed.
- (j) Suggest how the proteins coded for by *BVDV-CP* may lead to BYDV resistance. [2]
1. They may compete with the BYDV virus for surface glycoproteins that allow entry into host cells/ bind to nucleocapsid and prevent entry
 2. They may directly bind with viral capsid and break the virus down
 3. AVP (showing how the specific proteins may interfere with a specific point in the viral life cycle but should not cause nonspecific harm to host)

[Total: 23]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

Question 3

(a) Describe the principles and processes of the Polymerase Chain Reaction (PCR). [10]

Step 1: Denaturation

1. First, an excess of these primers are mixed with the DNA fragment to be amplified.
2. Primers are single stranded DNA sequences that are complementary to the sequence of DNA being amplified.
3. Primers are required for DNA replication as DNA polymerase needs an existing free 3' hydroxyl end for chain extension to take place.
4. This mixture of primer and fragment is heated to about 95°C for 30 sec.
5. At this temperature, the double-stranded DNA fragment dissociates into single strands. Due to the breakage of hydrogen bonds between complementary bases.

Step 2: Annealing of Primers

6. Next, the solution is allowed to cool to about 55- 68°C for 30 sec.
7. each strand of the fragment base-pairs with a complementary primer flanking the region to be amplified, (idea of annealing)
8. complementary base pairing

Step 3: Primer Extension

9. A thermo-stable type of DNA polymerase,
10. called Taq polymerase is required, along with a supply of all four nucleotides.
11. Polymerisation is usually carried out at 70-72°C for 1.5 minutes (optimum temperature for enzyme).
12. Polymerase copies the rest of the fragment as if it were replicating DNA, by adding nucleotides to the 3' OH of primer/growing strand
13. Because both DNA strands are replicated, there are now two copies of the original fragment.
14. Steps 1 to 3 are now repeated. For 25-30 cycles.

Max 10

(b) Discuss how photosynthesis plays a critical role in sustaining life on earth.

You should consider both specific processes in photosynthesis that sustain life and its role in reducing the impact of climate change. [15]

1. Specific Processes in photosynthesis

- State photolysis of water
- Water is split by light into hydrogen ions, electrons and oxygen
- Oxygen released used by other organisms in aerobic respiration to produce ATP for energy (Idea on how life is sustained)
- State calvin cycle
- Production of carbohydrates/starch
- Via reduction of phosphoglyceric acid (PGA/GP) to Triose Phosphate (TP)
- Carbohydrates used by other consumers in food chain to produce energy in the form of ATP
- Via cellular respiration

2. Role in reducing impact of climate change

- As carbon sinks that remove carbon dioxide from the atmosphere
- Idea of reducing greenhouse effect as carbon dioxide is a greenhouse gas
- Carbon dioxide fixation in calvin cycle
- Carbon dioxide from the atmosphere combines with RuBP to form GP.

3. Description of carbon sinks

- A carbon sink refers to a natural or artificial reservoir that accumulates and stores some carbon-containing chemical compound for an indefinite period.
- The process by which carbon sinks remove carbon dioxide from the atmosphere is known as carbon sequestration.
- Involves forests
- Involves plankton in oceans
- Involves northern hemisphere regrowths

QWC – Answer clearly focuses on role of photosynthesis BOTH in sustaining life and reducing the impact of climate change

Question 4

(a) Describe the principles and processes of Southern Blotting. [10]

1. Southern Blotting involves Nucleic Acid Hybridisation
2. Objective is to highlight specific nucleic acid sequences by using radioactive or fluorescently labelled probes (Idea)
3. DNA from the sample is cleaved into restriction fragments with a restriction endonuclease, and
4. fragments are separated by gel electrophoresis.
5. The gel slab containing the DNA fragment is placed under a nitrocellulose membrane and a stack of paper towels.
6. These are placed on top of a sponge in a tray of alkaline solution.
7. The absorbent paper towels draw the solution towards themselves. This capillary action draws the alkali solution through the gel.
8. denaturing the double stranded DNA fragments.
9. The single stranded DNA on the gel is drawn upwards to the nitrocellulose membrane.
10. Next, a probe corresponding/complementary to a specific nucleotide sequence is poured over the nitrocellulose membrane.
11. Probe needs to be single-stranded DNA/RNA; Probe needs to be labelled
12. After hybridization, the membrane is washed to remove any unhybridized probes.
13. Any fragment that has a nucleotide sequence complementary to the probe's sequence will hybridize (base-pair) with the probe.
14. If the probe has been radioactively labeled, and the sheet will show a band of radioactivity where -the probe hybridized with the complementary fragment when autoradiography is performed.

Max 10

- (b) It can be argued that humans have changed the levels of global respiration via economic activities and this has contributed to climate change.

Explain the processes in cellular respiration that release carbon dioxide and discuss how these and other human factors can contribute to climate change. [15]

1: Processes in cellular respiration that release carbon dioxide

- Link Reaction & Krebs Cycle in aerobic respiration
- Occurs in the matrix of mitochondria
- 6 carbon dioxide molecules released per glucose
- Via Oxidative Decarboxylation
- Anaerobic respiration in yeast
- When pyruvate converted to ethanol
- 2 carbon dioxide released per glucose

Max 5

2. How these processes lead to climate change

- Increased aerobic respiration rates lead to increased release of carbon dioxide into atmosphere (idea) / accept reverse
- Increased respiration rates from livestock farming
- Increased respiration rates from crop farming
- Increased respiration rates in oceans due to increased growth of marine organisms due to increased pollution (or other relevant causes) / idea of eutrophication
- Further idea of eutrophication blocking sunlight and thus leading to death and decomposition of organic plant matter in water bodies – also releasing carbon dioxide into atmosphere.
- Increased respiration rates via decomposition of organic matter by microbes (due to deforestation/clearing land for agriculture)

Max 5

3. Brief description of how these lead to global warming

- Due to increased release of greenhouse gasses such as carbon dioxide
- Greenhouse effect – trapping of long wave radiation in the atmosphere

4. Other human factors leading to climate change

- Livestock farming – impacts other than increased respiration
- Brief description (linking to climate change)
- Deforestation
- Brief description (linking to climate change)
- Burning of fossil fuels
- Brief description (linking to climate change)

Max 4

QWC – At least one point from each Section 3 and two points from each of the other sections.